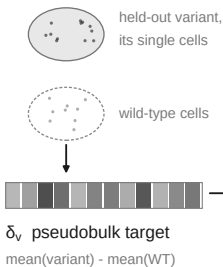


1 Measurement



2 Model

variant features (θ / ESM)



not the variant identity,
not its cells

feature to response
model

$\hat{\delta}_v$ predicted profile



3 Evaluation

Direction
(Pearson- δ)

Discrimination (PDS)

is it nearest its own allele
among its siblings?

both axes defined in e

4 Verdict

- measurable?
- benchmarkable?
- worth modelling?

expand / redesign / exclude

5 Resolution diagnostic (single cell)

same single cells, no averaging

detection: variant vs wild type

identification: variant vs siblings



can the ground truth
reward an allele model?